

PROJECT REPORT

Hidden Market States

Financial Market Regime Detection Using Hidden Markov Models

with Baum–Welch Parameter Learning and Viterbi Decoding

A Probabilistic Sequence-Modelling Approach to Detecting Bull, Bear, and Sideways Market Regimes from Observable Stock-Market Behaviour

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Date: August 2026

Keywords: Hidden Markov Model, Baum–Welch, Viterbi, Market Regime Detection, Time-Series Analysis, Quantitative Finance, S&P 500

Abstract

Financial markets are commonly treated as noisy, unstructured sequences of price observations. This project proposes an alternative view: that the apparent complexity of market behaviour can, to a meaningful extent, be explained by a small number of latent behavioural regimes — conventionally described as Bull, Bear, and Sideways markets. We model the market as a Hidden Markov Model (HMM) in which observable quantities (daily returns, rolling volatility, trading volume, and momentum) are generated by an underlying, unobserved regime that evolves over time according to Markovian dynamics.

The Baum–Welch algorithm (a variant of Expectation–Maximization built on the Forward–Backward procedure) is used to learn the model's initial-state distribution, transition matrix, and emission parameters directly from historical market data, without requiring any hand-labelled regime annotations. The Viterbi algorithm is then applied to decode the single most probable sequence of hidden regimes underlying the observed data. The resulting states are interpreted financially — as Bull, Bear, or Sideways — based on their learned statistical signatures (mean return and volatility), rather than being assumed in advance.

The project further analyses regime persistence, regime-to-regime transition probabilities,

expected regime duration, and short-horizon regime forecasting, and evaluates the model using log-likelihood, AIC/BIC, and qualitative validation against known historical market events. The deliverable is implemented in two layers: a from-scratch NumPy implementation of Forward–Backward, Baum–Welch, and Viterbi to demonstrate mathematical understanding, and a practical Gaussian-emission HMM applied to a real market index, presented through an interactive analytical dashboard.

1. Introduction

1.1 Background

Stock markets are widely believed to move through distinct behavioural phases — sustained upward growth (Bull), sustained decline (Bear), and range-bound, low-conviction movement (Sideways). Traders, analysts, and institutions routinely refer to "the current regime" of the market, yet this regime is never directly observed. What is observed is a stream of prices, returns, volatilities, and traded volumes; the regime itself must be inferred.

This inference problem — recovering an unobserved state from a sequence of observed, probabilistically related emissions — is precisely the structure that a Hidden Markov Model (HMM) is designed to solve. HMMs have a long history in speech recognition, bioinformatics, and, increasingly, quantitative finance, where they are used to model regime-switching behaviour in asset returns.

1.2 Motivation

Most introductory machine-learning projects in finance default to price-prediction using deep sequence models (LSTM, Transformer architectures). This project deliberately takes a different, more classical route: rather than predicting a price, it models the market as a stochastic process governed by a small number of latent states. This framing is mathematically transparent, computationally lightweight, and — critically — interpretable: every parameter the model learns has a direct financial meaning (a mean return, a volatility level, a persistence probability), which makes the project defensible as both an academic exercise in probabilistic modelling and a piece of applied financial analysis.

2. Problem Statement

Observable market variables (returns, volatility, volume) are generated by an underlying, unobservable market regime. Standard time-series techniques treat the sequence as either fully

random or governed by a single global process, and therefore fail to capture the fact that market behaviour changes character over time in a structured, regime-dependent way.

The problem this project addresses is twofold:

1. Given historical market observations, learn a statistical model of how market regimes evolve and how each regime generates observable behaviour, without pre-labelled regime data.
2. Given the learned model and an observed sequence, recover the most probable historical sequence of hidden regimes, interpret those regimes financially, and characterise their persistence and transition dynamics.

3. Research Question and Hypothesis

3.1 Research Question

Can Hidden Markov Models infer meaningful financial market regimes from observable stock-market behaviour, and can the learned transition dynamics be used to characterise regime persistence and transitions?

3.2 Hypothesis

Financial markets exhibit temporal patterns that can be represented by a small number of latent regimes, and an HMM trained using the Baum–Welch algorithm can learn these regimes directly from market features, while Viterbi decoding can recover their most probable historical sequence.

4. Objectives

- To engineer a set of observable market features (return, volatility, volume, momentum) suitable for regime modelling.
- To implement the Forward, Backward, Baum–Welch, and Viterbi algorithms from first principles, in order to demonstrate mathematical and algorithmic understanding.
- To train a Gaussian-emission HMM on real historical index data (specifically the S&P 500) and learn its transition and emission parameters.
- To decode the most probable historical regime sequence using Viterbi and interpret the discovered states as Bull, Bear, or Sideways based on their statistical signatures.
- To analyse regime persistence, regime duration, and regime-to-regime transition probabilities.

- To evaluate the model rigorously using log-likelihood, AIC/BIC, and historical event validation, rather than a classification-accuracy metric that assumes ground-truth labels which do not exist.
- To present the results through a clear, interactive visual dashboard.

5. Theoretical Background

5.1 Hidden Markov Models

A Hidden Markov Model describes a system that transitions between a finite set of hidden states according to Markovian dynamics — that is, the probability of the next state depends only on the current state, not on the full history. At each time step, the system in a given hidden state produces an observable output according to a state-specific emission distribution. The hidden state itself is never directly observed; only the emissions are.

Formally, an HMM is defined by the triple $\lambda = (A, B, \pi)$:

- π — the initial-state distribution, $\pi_i = P(S_1 = i)$
- A — the state-transition matrix, $A_{ij} = P(S_t = j \mid S_{t-1} = i)$
- B — the emission model, $B_j(x) = P(X_t = x \mid S_t = j)$

In this project, the hidden states $\{S_1, S_2, S_3\}$ correspond to latent market regimes (Bull, Bear, Sideways), and the observations X_t are the engineered market features at time t .

5.2 Forward–Backward Algorithm

The Forward algorithm computes, for every time step and every state, the joint probability of the observations seen so far and being in that state at that time, using a dynamic-programming recursion that avoids the exponential cost of enumerating every possible state path. The Backward algorithm computes the complementary quantity: the probability of all future observations given the current state. Together, the Forward and Backward passes allow the exact probability of being in any state at any time, given the entire observation sequence, to be computed — this is the statistical foundation on which Baum–Welch is built.

5.3 Baum–Welch Algorithm (Learning)

Baum–Welch is a specific instance of the Expectation–Maximization (EM) algorithm applied to HMMs. It answers the question: given a sequence of observations, what are the model parameters (π, A, B) most likely to have generated them? It proceeds iteratively:

- **E-step:** Using the current parameter estimates, run Forward–Backward to compute the

posterior probability of each state at each time step, and of each state transition at each time step.

- **M-step:** Re-estimate π , A, and B using these posterior probabilities as soft weights.

Repeat until the log-likelihood of the observed data converges. No labelled regime data is required — the algorithm discovers statistically distinct hidden states purely from the structure of the observations.

5.4 Viterbi Algorithm (Decoding)

Once the model parameters have been learned, the Viterbi algorithm finds the single most probable sequence of hidden states given the full observation sequence — as opposed to Forward–Backward, which gives the probability of each state independently at each time step. Viterbi uses a dynamic-programming recursion over the trellis of states and time steps, tracking the highest-probability path to each state at each time and backtracking once the end of the sequence is reached. This produces the final regime timeline used throughout the rest of the analysis.

5.5 A Note on Interpretation

It is important to state explicitly that an HMM does not inherently know what "Bull" or "Bear" means. Baum–Welch discovers statistically distinct hidden states based purely on patterns in the data; the financial labels are assigned afterward, by examining each learned state's mean return and volatility and mapping the state with the highest positive mean return to "Bull", the state with the most negative mean return and highest volatility to "Bear", and the remaining state to "Sideways". This distinction between statistical discovery and financial interpretation is a central methodological point of the project.

6. Methodology

6.1 Data Source

The project uses a single major market index as its primary dataset, rather than a heterogeneous basket of individual stocks, because the objective is to characterise the regime of the market as a whole rather than the idiosyncratic behaviour of any one company. **For the purposes of this project, the S&P 500 index has been selected as the definitive dataset.** Serving as the benchmark index of the US equity market, it provides a robust, broad, and internationally recognised context for modelling regime behaviour.

6.2 Feature Engineering

Raw closing price is not used directly as the model observation. Instead, four derived features are engineered to capture different aspects of market behaviour:

Feature	Formula / Definition	Financial Interpretation
Daily return r_t	$r_t = \ln(P_t / P_{t-1})$	Direction and magnitude of price movement
Rolling volatility σ_t	$\sigma_t = \text{Std}(r_{t-W+1}, \dots, r_t)$	Degree of market turbulence over a rolling window
Trading volume	Raw or normalised traded volume	Level of market participation / activity
Momentum	$\text{Momentum}_t = P_t - P_{t-W}$ (or normalised)	Sustained directional pressure over a window

6.3 Data Preprocessing

1. Retrieve historical OHLCV data for the S&P 500 and sort chronologically.
2. Handle missing values and non-trading-day gaps.
3. Compute log returns, rolling volatility, volume features, and momentum.
4. Standardise continuous features: $X' = (X - \mu) / \sigma$, to prevent scale dominance across features of different magnitude.
5. Assemble the final multivariate observation matrix used as HMM input.

6.4 Modelling Approach: Discrete vs. Gaussian Emissions

Financial features such as return and volatility are inherently continuous, whereas a classical HMM formulation is often introduced using discrete observation symbols. Two modelling approaches are considered:

- **Approach A — Discretised HMM:** Continuous observations are binned into categorical levels (e.g. Return: Negative / Neutral / Positive). This approach is conceptually simple

and useful for explaining the mechanics of the model.

- **Approach B — Gaussian-Emission HMM (adopted for the primary model):**
Continuous observations are modelled directly, with each hidden state i associated with a multivariate Gaussian emission distribution: $X_t | S_t = i \sim N(\mu_i, \Sigma_i)$. This preserves the full information content of the features and yields a richer, more defensible financial interpretation of each regime.

6.5 Two-Layer Implementation Strategy

The project is deliberately implemented in two layers, to demonstrate both theoretical rigour and practical competence:

- **Layer 1 — Educational implementation:** Forward, Backward, Baum–Welch, and Viterbi implemented from first principles in NumPy, validated against a small synthetic example with known parameters.
- **Layer 2 — Practical model:** A Gaussian-emission HMM applied to real historical S&P 500 index data for the final analysis and dashboard.

7. System Architecture

The end-to-end pipeline is organised as follows:





Evaluation & Interactive Dashboard

8. Implementation Plan

Phase	Description	Key Output
1. Data Acquisition	Retrieve and clean historical OHLCV data for the S&P 500	Clean, sorted time-series dataset
2. Exploratory Analysis	Plot price, returns, volatility; inspect distributions and correlations	Confirmation of distinct behavioural periods
3. Preprocessing	Compute features; standardise continuous variables	Final observation matrix
4. Model Initialisation	Select number of states ($k = 3$); initialise π , A , μ , Σ	Initialised HMM object
5. Baum–Welch Training	Iterate Forward–Backward and parameter re-estimation to convergence	Learned transition matrix A and emission parameters
6. Viterbi Decoding	Decode the most probable hidden-state sequence	Regime timeline
7. State Interpretation	Compute per-state mean return, volatility, volume, duration	Bull / Sideways / Bear labelling
8. Financial Analysis	Regime duration, transition probabilities, next-regime	Transition diagram, duration statistics

Phase	Description	Key Output
	forecasting	
9. Evaluation	Log-likelihood, AIC/BIC across $k = 2 \dots 5$ states; historical validation	Model selection justification
10. Dashboard	Interactive presentation of all results	Final deliverable notebook / application

8.1 Train / Test Split

Because market data is a time series, the train/test split must preserve temporal order — the data is never randomly shuffled. An earlier contiguous block of history is used for training (parameter learning), and a later, held-out contiguous block is used for testing (out-of-sample regime decoding and evaluation).

9. Evaluation Methodology

Because hidden regimes have no ground-truth labels, evaluation cannot rely on classification accuracy. Instead, the model is evaluated along five complementary dimensions:

- **Log-likelihood, $\log P(\mathbf{X} \mid \lambda)$:** How well the trained model explains the observed data.
- **AIC / BIC:** Trade-off between model fit and complexity, used to justify the chosen number of states (k).
- **Financial interpretability:** Whether discovered states show clearly distinct mean return, volatility, and volume.
- **Regime stability:** Whether the decoded sequence produces persistent regimes rather than implausible day-to-day switching.
- **Historical event validation:** Whether known downturns / crashes align with periods decoded as high-volatility Bear regimes.

10. Expected Results and Analysis

10.1 Transition Matrix

Baum–Welch is expected to learn a transition matrix exhibiting strong diagonal dominance — i.e., high self-transition probabilities — reflecting the well-documented persistence of market regimes.

10.2 Emission Characteristics

Learned emission characteristics per regime will map highest positive mean return to Bull, most negative mean return and highest volatility to Bear, and low-conviction parameters to Sideways.

10.3 Regime Duration

For a state i with self-transition probability p_{ii} , the expected regime duration under a geometric persistence assumption is approximately $E[D_i] = 1 / (1 - p_{ii})$. This gives a mathematically grounded estimate of how long the market is expected to remain in each regime.

10.4 Regime Transition Analysis

Beyond the raw transition matrix, the project examines specific transition pathways (e.g. Bull → Sideways → Bear versus direct Bull → Bear transitions).

10.5 Short-Horizon Regime Forecasting

The learned transition matrix can be used to estimate the probability distribution over the next period's regime given the current regime, $P(S_{t+1} | S_t)$.

11. Dashboard and Presentation Layer

The final deliverable is presented as a polished notebook or lightweight interactive application (e.g. Streamlit), organised into the following sections:

- Market overview — current decoded regime and associated confidence.
- Price chart annotated with the decoded regime timeline.
- Feature analysis — return, volatility, and volume plots.
- Transition matrix heatmap.
- Regime characteristics summary (return / volatility / duration per regime).

- Full Viterbi-decoded historical regime timeline.
- Next-regime probability panel.

12. Technology Stack

- **Core language:** Python
- **Numerical computing:** NumPy, SciPy
- **Data handling:** Pandas
- **Market data source:** yfinance (targeting S&P 500)
- **Custom HMM (Layer 1):** Pure Python + NumPy implementation
- **Visualisation:** Matplotlib, Seaborn
- **Interactive application:** Streamlit

13. Scope and Limitations

- The model assumes a fixed, small number of hidden states (typically $k = 3$).
- The first-order Markov assumption is a simplification; real regime persistence may depend on longer history.
- Regime labels (Bull / Bear / Sideways) are assigned post-hoc based on learned statistics, without universally agreed ground truth.
- The project models a single index (S&P 500); cross-asset or multi-market generalisation is left as future work.

14. Future Work

- Extend the model to a multi-asset or multi-index setting to study regime co-movement across markets.
- Explore higher-order or hierarchical HMM variants.
- Incorporate macroeconomic indicators (interest rates, inflation prints).
- Compare regime-detection performance against Markov-Switching GARCH models.

15. Conclusion

This project models the stock market as a Hidden Markov Model in which observable features — returns, volatility, volume, and momentum — are generated by an underlying, unobserved market regime. The Baum–Welch algorithm is used to learn the model's transition and emission

parameters directly from historical data, and the Viterbi algorithm is used to decode the most probable historical sequence of hidden regimes. The resulting states are interpreted as Bull, Bear, and Sideways markets based on their learned statistical signatures, and the model's transition dynamics are used to analyse regime persistence, regime transitions, and short-horizon regime forecasting.

By keeping the modelling approach classical and mathematically transparent rather than relying on opaque deep-learning architectures, the project delivers a result that is simultaneously rigorous, interpretable, and directly tied to a genuine financial question: whether the apparent complexity of market behaviour can be explained by a small number of hidden behavioural states.

References

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Appendix A — Mathematical Notation

Symbol	Meaning
S_1, S_2, S_3	Hidden market regimes (Bull, Bear, Sideways)
X_t	Observation vector at time t (return, volatility, volume, momentum)
π	Initial-state probability distribution

Symbol	Meaning
A	State-transition probability matrix
B	Emission probability model
$\lambda = (A, B, \pi)$	Full HMM parameter set
r_t	Log return at time t
σ_t	Rolling volatility at time t
μ_i, Σ_i	Mean vector and covariance matrix of the Gaussian emission for state i
$E[D_i]$	Expected duration of regime i